ELEC4150/8150 Electronic Design Proficiency

Project 2 – Demonstration & Defence

GENERAL GUIDELINES

Assessment of Project 2 will take place on 14th Sep 2026, the Monday of Week 8, during the practical class. Each student will be assigned a 30-min slot, to be scheduled and announced in advance. Students are encouraged to arrive earlier than the scheduled time to set up and test their systems.

Keep in mind that the assessment is not just about the completion of the project, but also your understanding and explanation. More specifically we are looking for:

- Result - what is completed and how does it work according to the project specification.
- Understanding - any theoretical and technical background, why you chose the solution, advantages and disadvantages of the design, as well as reasons why your design does not work if it doesn't.
- Explanation and demonstration - how clearly and concisely you can explain your solution, including functions of individual components and sections.
- Others – such as neatness and organisation for demonstration of the solution.

ASSESSMENT RUBRIC

Demonstration and Defence of Project 2 accounts for 25 marks of the unit, with 0.5 mark increments. An extra of 3 marks is attributed to the bonus part, but the total mark you can receive will be capped at 25 marks. For effective time management, it is recommended your selected piece of audio signal should be trimmed to less than 30 seconds containing both vocal and acoustic components.

Part 1 (7 marks)

- Handling of audio signal (3 marks)
  - ➢ Import the audio signal and display it in the time domain
  - ➢ Output the audio via speaker/headphone, and display its real-time spectrogram
  - ➢ Demonstrate sampling rate conversion and answer relevant questions
- Vocal extraction (2 marks)
  - ➢ Demonstrate your extracted vocal component, including the spectrum/spectrogram

  - Explain your source separation method and answer relevant questions
- Karaoke effect (2 marks)
  - Explain your rationale behind how to quantify the attenuation of vocal component
  - Demonstrate the enhanced acoustic component and answer relevant questions

Part 2 (7 marks)

- Compression (1 mark)
  - Explain your signal compression and demonstrate it meets the specification
- Modulation (2 marks)
  - Explain your modulation method and answer relevant questions
  - Illustrate the spectrum of the modulated signal with respect to specification
- Demodulation (1 mark)
  - Explain your demodulation method and demonstrate the demodulated signal
- Noise effect and countermeasure (3 marks)
  - Simulate additive white Gaussian noise upon transmitted signal with specified SNR
  - Explain your countermeasures against the noise and answer relevant questions
  - Demonstrate the effect of your countermeasures and evaluate the performance

Part 3 (7 marks)

- Model system (2 marks)
  - Explain your code and answer relevant questions
  - Illustrate the outputs along with the inputs
- Delay simulation (2 marks)
  - Explain the corresponding code and show the results
  - Demonstrate how sub-sample delay resolution is achieved
- Sound source localisation (3 marks)
  - Explain your method and answer relevant questions
  - Demonstrate your result
  - Evaluate the accuracy of your method

Others (4 marks)

- Rationale behind choice of methods, incl. consideration on complexity/speed (2 marks)
- Testing and troubleshooting capabilities in your design (1 mark)
- Neatness and organisation of your software (1 mark)

Bonus Part (3 marks)

- Music note transcriber (1 mark)
  - Demonstrate your system and explain how it works